

Name: _____

Roll number: _____

Time Allowed: 15 min

Total Marks: 20

Question#1 (10 Marks)----- CLO3

Write code to store the elements of array x in each column of array y. You are not required to display the output. The expected results are shown in the box for your understanding only. Do not hardcode values; you are required to use a nested loop.

```
int x[5] = {23, 46, 78, 12, 56};  
int y[5][4];
```

```
for (int i=0 ; i<5 ; i++)  
{  
    for(int j=0 ; j<4 ; j++)  
    {  
        y[i][j]=x[i];  
    }  
}
```

23	23	23	23
46	46	46	46
78	78	78	78
12	12	12	12
56	56	56	56

Question#2 (10 Marks)----- CLO3

Write a function, which finds a zero in 2-D array and replaces the entire row and column with zeros of the received arrays.

Sample Input Array:

23	45	-78	45
4	67	89	90
6	7	0	11
13	12	15	14

Sample Output/changed Array:

23	45	0	45
4	67	0	90
0	0	0	0
13	12	0	14

```
int zeroRow = -1, zeroCol = -1;  
  
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < cols; j++) {  
        if (arr[i][j] == 0) {  
            zeroRow = i;  
            zeroCol = j;  
            break;  
        }  
    }  
    if (zeroRow != -1)  
        break;  
}  
  
for (int j = 0; j < cols; j++)  
    arr[zeroRow][j] = 0;  
for (int i = 0; i < rows; i++)  
    arr[i][zeroCol] = 0;
```